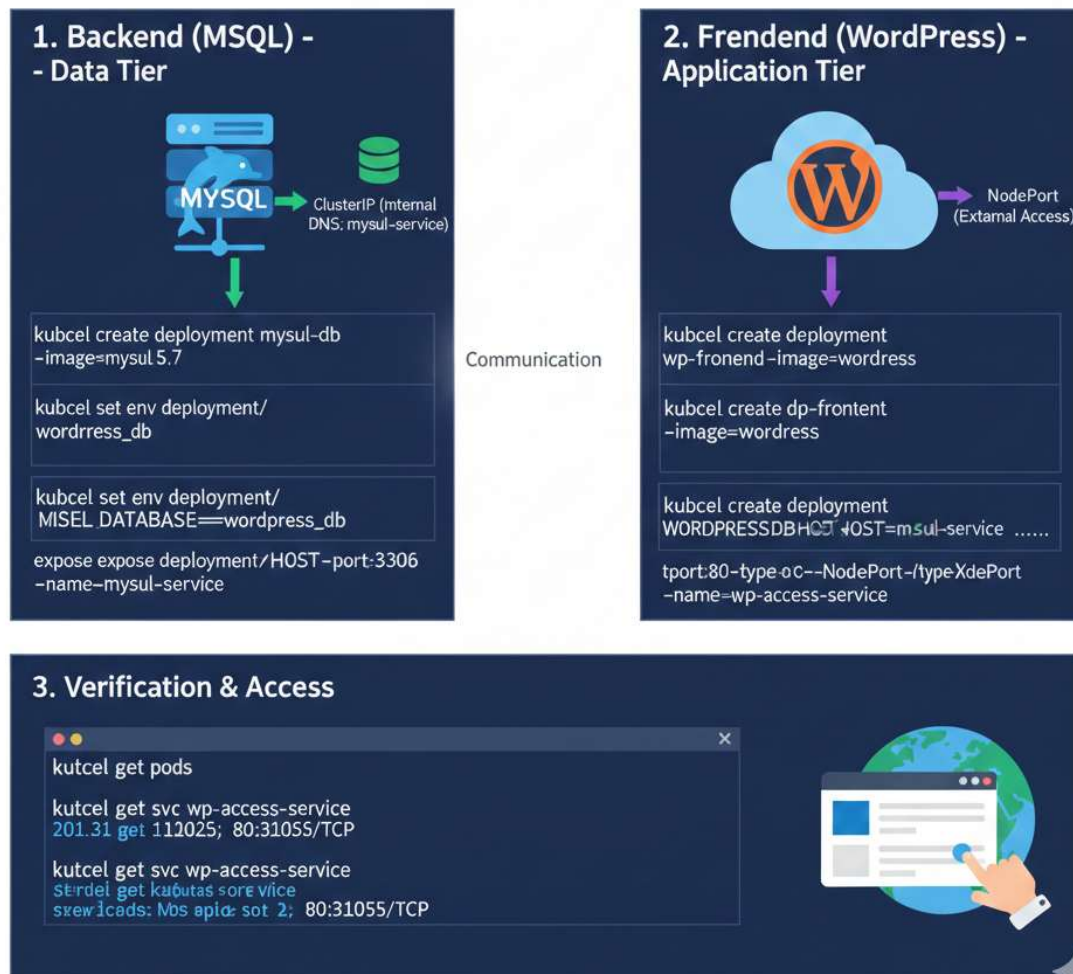


Deploying WordPress & MySQL on Kubernetes

To deploy a functional multi-tier web application where the Frontend (WordPress) communicates with a Backend Database (MySQL) using Kubernetes commands without pre-written YAML files.



Deploy the Backend (MySQL)

First, create the database engine and expose it internally so other pods can find it by name.

Create the deployment

```
controlplane:~$ kubectl create deployment mysql-db --image=mysql:5.7
deployment.apps/mysql-db created
```

```
controlplane:~$ kubectl get pod -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE   NOMINATED NODE   READINESS GATES
mysql-db-6c8f9c7ff4-prspr           1/1     Running   0           4m55s  192.168.1.5   node01  <none>           <none>
```

Configure database credentials via environment variables

```
controlplane:~$ kubectl set env deployment/mysql-db MYSQL_ROOT_PASSWORD=shiva-pass MYSQL_DATABASE=wordpress_db
deployment.apps/mysql-db env updated
```

Expose the deployment internally

Creates a DNS entry 'mysql-service' that the frontend will use

```
controlplane:~$ kubectl expose deployment mysql-db --port=3306 --name=mysql-service
service/mysql-service exposed
```

```
controlplane:~$ kubectl describe svc mysql-service
Name:                                mysql-service
Namespace:                           default
Labels:                               app=mysql-db
Annotations:                           <none>
Selector:                             app=mysql-db
Type:                                  ClusterIP
IP Family Policy:                      SingleStack
IP Families:                           IPv4
IP:                                    10.105.108.159
IPs:                                   10.105.108.159
Port:                                  <unset> 3306/TCP
TargetPort:                            3306/TCP
Endpoints:                             192.168.1.5:3306
Session Affinity:                      None
Internal Traffic Policy:                Cluster
Events:                                 <none>
```

Deploy the Frontend (WordPress)

Next, deploy the WordPress application and point it to the MySQL service name created in Step 1.

Create the deployment

```
controlplane:~$ kubectl create deployment wp-frontend --image=wordpress
deployment.apps/wp-frontend created
controlplane:~$ kubectl get pod -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE   NOMINATED NODE   READINESS GATES
mysql-db-6c8f9c7ff4-prspr           1/1     Running   0           14m   192.168.1.5   node01  <none>           <none>
wp-frontend-8597487756-9cv94        1/1     Running   0           28s   192.168.1.6   node01  <none>           <none>
```

Connect WordPress to the Backend service

Note: `WORDPRESS_DB_HOST` must match the service name from Step 1

```
controlplane:~$ kubectl set env deployment/wp-frontend WORDPRESS_DB_HOST=mysql-service WORDPRESS_DB_NAME=wordpress_db WORDPRESS_DB_USER=root WORDPRESS_DB_PASSWORD=shiva-pass
deployment.apps/wp-frontend env updated
controlplane:~$
```

Expose WordPress to external traffic

```
controlplane:~$ kubectl expose deployment wp-frontend --port=80 --type=NodePort --name=wp-access-service
service/wp-access-service exposed
controlplane:~$ kubectl describe svc wp-access-service
Name:                wp-access-service
Namespace:            default
Labels:               app=wp-frontend
Annotations:          <none>
Selector:             app=wp-frontend
Type:                 NodePort
IP Family Policy:     SingleStack
IP Families:          IPv4
IP:                   10.96.219.171
IPs:                  10.96.219.171
Port:                 <unset> 80/TCP
TargetPort:           80/TCP
NodePort:             <unset> 30939/TCP
Endpoints:            192.168.1.7:80
Session Affinity:     None
External Traffic Policy: Cluster
Internal Traffic Policy: Cluster
Events:               <none>
```

Verification

Verify that both components are running and healthy.

```
controlplane:~$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
mysql-db-6c8f9c7ff4-prspr          1/1     Running   0           26m
wp-frontend-8447c56f57-t8sqm       1/1     Running   0           9m14s
controlplane:~$ kubectl get svc wp-access-service
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
wp-access-service   NodePort    10.96.219.171 <none>        80:30939/TCP     2m13s
controlplane:~$
```

Database Shell: To enter the database manually

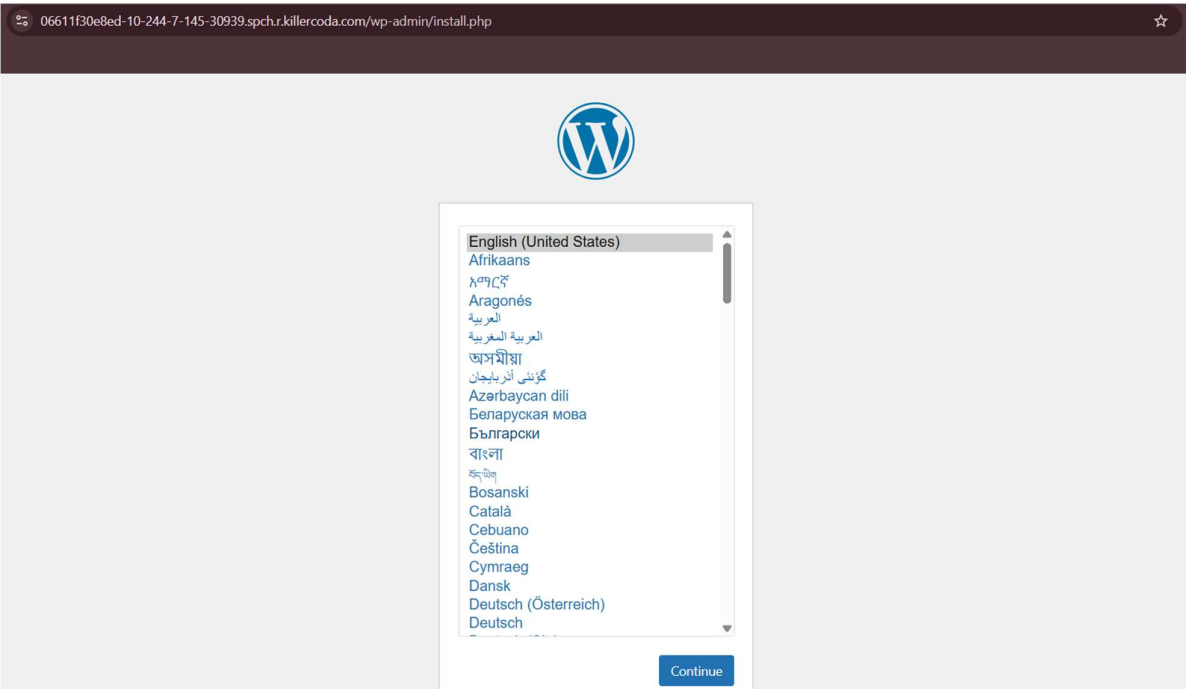
```
controlplane:~$ kubectl exec -it mysql-db-6c8f9c7ff4-prspr -- bash
bash-4.2# mysql -uroot -pshiva-pass
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 19
Server version: 5.7.44 MySQL Community Server (GPL)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
| wordpress_db |
+-----+
5 rows in set (0.00 sec)
```



Welcome

Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world.

Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title

Username

Usernames can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.

Password [Hide](#)

Important: You will need this password to log in. Please store it in a secure location.

Your Email

Double-check your email address before continuing.

Search engine visibility ☐ Discourage search engines from indexing this site
It is up to search engines to honor this request.

[Install WordPress](#)